PSTAT 126: Regression Analysis - Final Project

Group Project

PART 1: Data Description and Descriptive Statistics

Question 1: Select a random sample

```
# Load the dataset
diamonds_data <- read.csv("Diamonds Prices2022.csv")
set.seed(67)
# Select a random sample of 1000 observations
diamonds_sample <- diamonds_data[sample(nrow(diamonds_data), 1000), ]
head(diamonds_sample)
```

```
          X carat       cut color clarity depth table price    x    y    z
10667 10667  0.90   Premium     F     VS2  62.3    58  4838 6.16 6.14 3.83
41671 41671  0.40     Ideal     G     VS2  61.1    56  1240 4.74 4.78 2.91
37921 37921  0.23 Very Good     E    VVS1  63.3    57   485 3.88 3.95 2.48
46833 46833  0.55     Ideal     F     VS1  61.3    57  1809 5.25 5.28 3.23
38452 38452  0.31     Ideal     G      IF  62.0    54  1028 4.39 4.35 2.71
26528 26528  1.52   Premium     E     VS1  60.1    59 16187 7.49 7.45 4.49
```

Question 2: Describe all the variables

```
# Summary
summary(diamonds_sample)
```

```
       X             carat           cut               color
 Min.   :  164   Min.   :0.200   Length:1000        Length:1000
 1st Qu.:14818   1st Qu.:0.400   Class :character   Class :character
 Median :27848   Median :0.700   Mode  :character   Mode  :character
 Mean   :27659   Mean   :0.802
 3rd Qu.:40868   3rd Qu.:1.040
 Max.   :53913   Max.   :3.000
   clarity              depth           table           price
 Length:1000        Min.   :43.00   Min.   :52.00   Min.   :  361.0
 Class :character   1st Qu.:61.20   1st Qu.:56.00   1st Qu.:  981.5
 Mode  :character   Median :62.00   Median :57.00   Median : 2406.5
                    Mean   :61.83   Mean   :57.43   Mean   : 3997.9
                    3rd Qu.:62.60   3rd Qu.:59.00   3rd Qu.: 5373.8
                    Max.   :69.50   Max.   :67.00   Max.   :18806.0
       x               y               z
 Min.   :3.730   Min.   :3.710   Min.   :0.000
 1st Qu.:4.720   1st Qu.:4.730   1st Qu.:2.910
 Median :5.700   Median :5.710   Median :3.510
 Mean   :5.735   Mean   :5.737   Mean   :3.542
 3rd Qu.:6.522   3rd Qu.:6.550   3rd Qu.:4.030
 Max.   :9.380   Max.   :9.310   Max.   :5.410
```

```r
# Structure
str(diamonds_sample)
```

```
'data.frame':   1000 obs. of  11 variables:
 $ X      : int  10667 41671 37921 46833 38452 26528 22724 21072 41496 7358 ...
 $ carat  : num  0.9 0.4 0.23 0.55 0.31 1.52 1.51 1 0.41 0.56 ...
 $ cut    : chr  "Premium" "Ideal" "Very Good" "Ideal" ...
 $ color  : chr  "F" "G" "E" "F" ...
 $ clarity: chr  "VS2" "VS2" "VVS1" "VS1" ...
 $ depth  : num  62.3 61.1 63.3 61.3 62 60.1 61.6 63.9 59.9 62.4 ...
 $ table  : num  58 56 57 57 54 59 59 56 60 56 ...
 $ price  : int  4838 1240 485 1809 1028 16187 10739 9248 1233 4216 ...
 $ x      : num  6.16 4.74 3.88 5.25 4.39 7.49 7.32 6.31 4.8 5.24 ...
 $ y      : num  6.14 4.78 3.95 5.28 4.35 7.45 7.3 6.34 4.82 5.28 ...
 $ z      : num  3.83 2.91 2.48 3.23 2.71 4.49 4.5 4.04 2.88 3.28 ...
```

Variable Descriptions

Our sample contains the following variables:

Response Variable:

- price: Price of the diamond in US dollars ranging from $361 to $18806. This is our target variable for prediction.

Quantitative Predictor Variables:

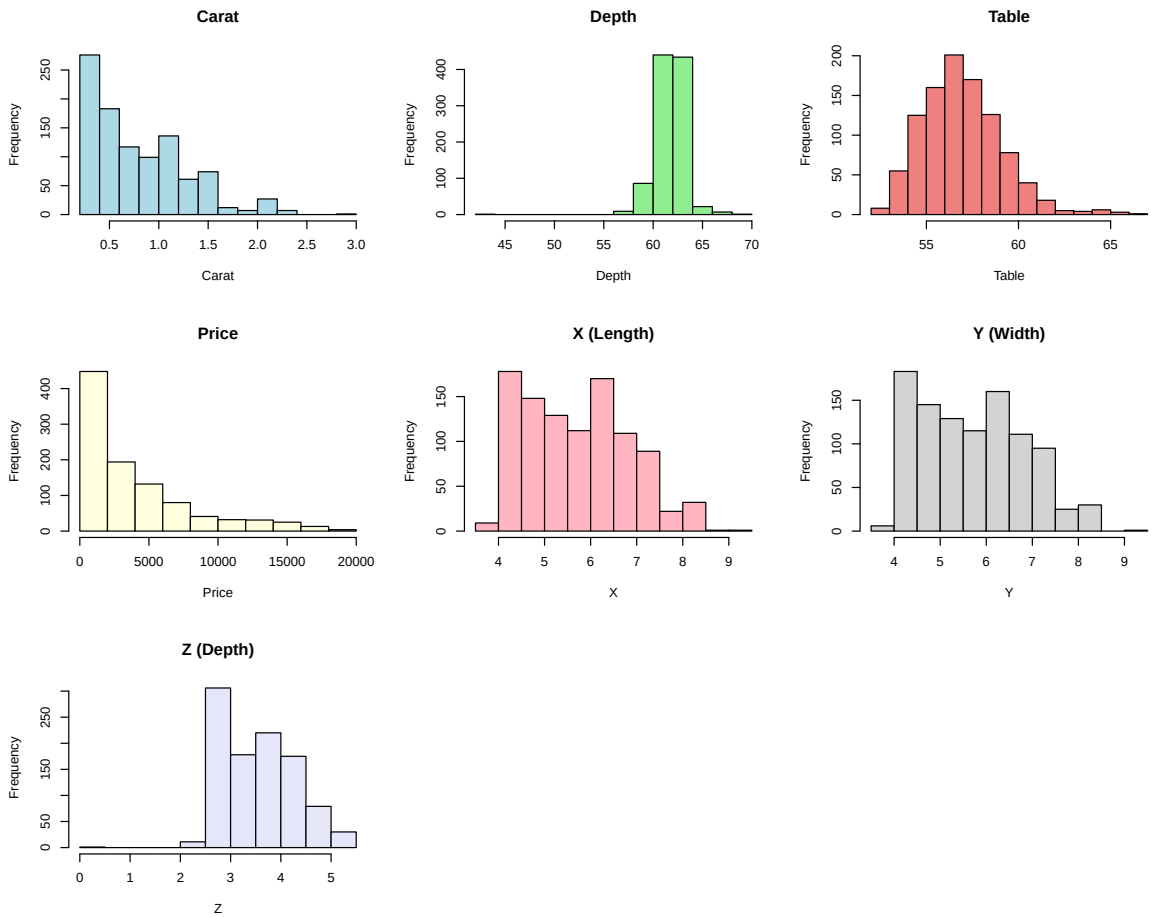
- carat: Weight of the diamond ranging from 0.2 to 3.0 carats.
- depth: Measures how deep the diamond is relative to its width ranging from 43% to 69.5%.
- table: Width of the top of the diamond relative to its widest point ranging from 52% to 67%.
- x: Length of the diamond in millimeters ranging from 3.73mm to 9.38mm.
- y: Width of the diamond in millimeters ranging from 3.71mm to 9.31mm.
- z: Depth of the diamond in millimeters ranging from 0mm to 5.41mm. (A measurement of 0mm is impossible which means that we have to remove some observations where measurements are 0mm.

Categorical Predictor Variables:

- cut: Quality of the diamond cut. The scale goes in order from worst to best (Fair, Good, Very Good, Premium, Ideal).
- color: Diamond color grade from worst to best (J to D).
- clarity: Measure of the diamonds internal flaws ranging from (I1 < SI2 < SI1 < VS2 < VVS2 < VVS1 < IF).

```r
# Histograms for continuous variables
par(mfrow = c(3, 3))

hist(diamonds_sample$carat, main = "Carat",
     xlab = "Carat", col = "lightblue")
hist(diamonds_sample$depth, main = "Depth",
     xlab = "Depth", col = "lightgreen")
hist(diamonds_sample$table, main = "Table",
     xlab = "Table", col = "lightcoral")
hist(diamonds_sample$price, main = "Price",
     xlab = "Price", col = "lightyellow")
hist(diamonds_sample$x, main = "X (Length)",
     xlab = "X", col = "lightpink")
hist(diamonds_sample$y, main = "Y (Width)",
```

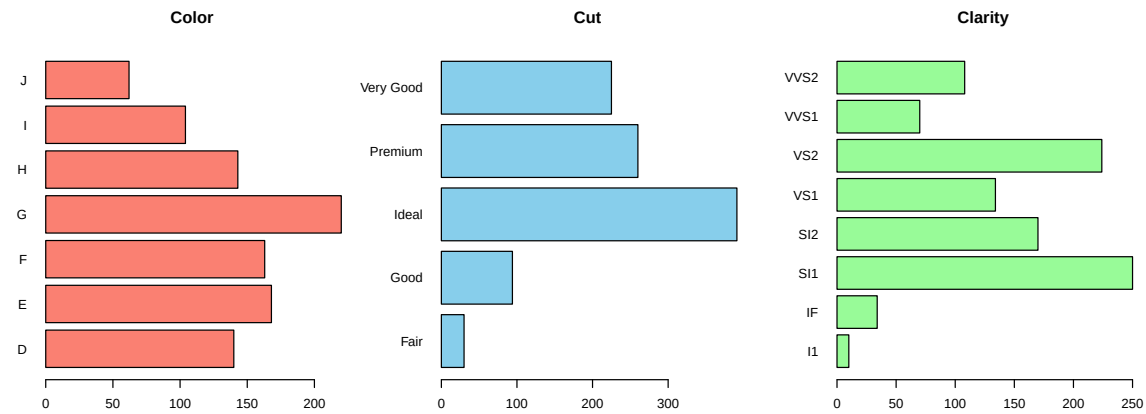
```
     xlab = "Y", col = "lightgray")
hist(diamonds_sample$z, main = "Z (Depth)",
     xlab = "Z", col = "lavender")
```



Comments on their Distributions: - Price and carat are right-skewed - Depth seems normal and stable - Table and X,Y,Z have high variability - X, Y, Z are slightly right-skewed.

```
# Bar plots for categorical variables
par(mfrow = c(1, 3))

barplot(table(diamonds_sample$color), main = "Color",
        col = "salmon", las=1, horiz=TRUE)
barplot(table(diamonds_sample$cut), main = "Cut",
        col = "skyblue", las=1, horiz=TRUE)
```

```r
barplot(table(diamonds_sample$clarity), main = "Clarity",
        col = "palegreen", las=1, horiz=TRUE)
```



Comments on Categorical Variables: Most diamonds have ideal or premium cuts. Distribution is fairly even across G, E, F, and H grades with fewer diamonds at the extremes (D and J) and I grades. SI1 and VS2 are the most common clarity grades, representing mid-range quality.

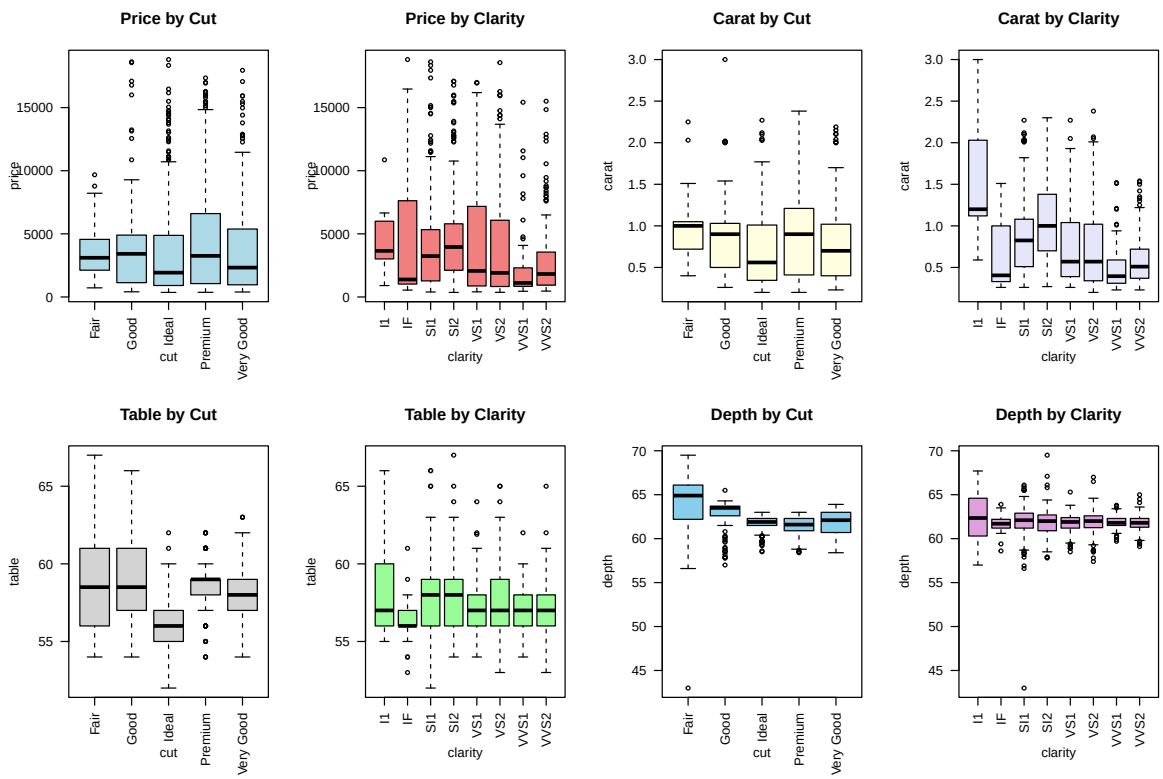
Question 3: Choose variables and determine correlation

```r
## Question 3: Choose variables and determine correlation

# Select 3 quantitative variables: table, carat, depth
# Select 2 categorical variables: cut, clarity

# 1. Correlation between QUANTITATIVE variables
quant_vars <- diamonds_sample[, c("price", "table", "carat", "depth")]
cor_matrix <- cor(quant_vars)
round(cor_matrix, 3)
```

```
       price  table carat  depth
price  1.000  0.117 0.913 -0.012
table  0.117  1.000 0.191 -0.256
carat  0.913  0.191 1.000  0.002
depth -0.012 -0.256 0.002  1.000
```

```r
# 2. Relationship between CATEGORICAL and QUANTITATIVE variables using ANOVA
# ANOVA tests if mean of quantitative variable differs across categorical groups
#Commenting out Summary to remove Clutter

# Test if mean price differs across cut categories
aov_price_cut <- aov(price ~ cut, data = diamonds_sample)
#summary(aov_price_cut)

# Test if mean price differs across clarity categories
aov_price_clarity <- aov(price ~ clarity, data = diamonds_sample)
#summary(aov_price_clarity)

# Test relationships for other quantitative variables
# Carat vs categorical variables
aov_carat_cut <- aov(carat ~ cut, data = diamonds_sample)
#summary(aov_carat_cut)

aov_carat_clarity <- aov(carat ~ clarity, data = diamonds_sample)
#summary(aov_carat_clarity)

# Table vs categorical variables
aov_table_cut <- aov(table ~ cut, data = diamonds_sample)
#summary(aov_table_cut)

aov_table_clarity <- aov(table ~ clarity, data = diamonds_sample)
#summary(aov_table_clarity)

# Depth vs categorical variables
aov_depth_cut <- aov(depth ~ cut, data = diamonds_sample)
#summary(aov_depth_cut)

aov_depth_clarity <- aov(depth ~ clarity, data = diamonds_sample)
#summary(aov_depth_clarity)

# Visualize relationships between categorical and quantitative variables
par(mfrow = c(3, 4))

# Price vs categorical variables
boxplot(price ~ cut, data = diamonds_sample,
        main = "Price by Cut", las = 2, col = "lightblue")
boxplot(price ~ clarity, data = diamonds_sample,
        main = "Price by Clarity", las = 2, col = "lightcoral")
```

```r
# Carat vs categorical variables
boxplot(carat ~ cut, data = diamonds_sample,
        main = "Carat by Cut", las = 2, col = "lightyellow")
boxplot(carat ~ clarity, data = diamonds_sample,
        main = "Carat by Clarity", las = 2, col = "lavender")

# Table vs categorical variables
boxplot(table ~ cut, data = diamonds_sample,
        main = "Table by Cut", las = 2, col = "lightgray")
boxplot(table ~ clarity, data = diamonds_sample,
        main = "Table by Clarity", las = 2, col = "palegreen")

# Depth vs categorical variables
boxplot(depth ~ cut, data = diamonds_sample,
        main = "Depth by Cut", las = 2, col = "skyblue")
boxplot(depth ~ clarity, data = diamonds_sample,
        main = "Depth by Clarity", las = 2, col = "plum")

# Interpretation of ANOVA results:
# If p-value < 0.05: There is a significant relationship (the means differ across groups)
# If p-value > 0.05: No significant relationship (the means are similar across groups)
```




```r
# 3 quantitative variables: table, carat, depth
# 2 categorical variables: cut, color

# Correlation matrix
quant_vars <- diamonds_sample[, c("price", "table", "carat", "depth")]
cor_matrix <- cor(quant_vars)
round(cor_matrix, 3)
```

```
       price  table carat  depth
```

```
price  1.000  0.117 0.913 -0.012
table  0.117  1.000 0.191 -0.256
carat  0.913  0.191 1.000  0.002
depth -0.012 -0.256 0.002  1.000
```

Correlation Interpretation:

- Price-Carat (r = 0.913): Very strong positive correlation. Carat weight is the strongest predictor of price.
- Price-Depth (r = -0.012): Negligible correlation. Depth percentage has little linear relationship with price.
- Price-Table (r = 0.117): Weak positive correlation. Table width has minimal impact on price.
- Carat-Depth/Table: Weak correlations.
- Depth-Table: Weak negative correlation.

ANOVA Results Summary: All categorical variables tested (cut, clarity) show statistically significant relationships with price, indicating that mean prices differ across quality categories.

Question 4: Multiple linear regression model

```r
model_initial <- lm(price ~ carat + cut + clarity + depth + table,
                    data = diamonds_sample)
summary(model_initial)
```

```
Call:
lm(formula = price ~ carat + cut + clarity + depth + table, data = diamonds_sample)

Residuals:
    Min      1Q  Median      3Q     Max 
-6665.2  -667.2   -93.8   498.3  8561.5 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -4863.65    2950.87  -1.648 0.099630 .  
carat         8527.96      97.67  87.318  < 2e-16 ***
```

```
cutGood        852.27     288.70   2.952 0.003231 **
cutIdeal       915.32     283.68   3.227 0.001294 **
cutPremium     929.44     275.08   3.379 0.000757 ***
cutVery Good   967.63     277.64   3.485 0.000513 ***
clarityIF     7059.50     504.28  13.999  < 2e-16 ***
claritySI1    4468.39     448.51   9.963  < 2e-16 ***
claritySI2    3629.65     451.30   8.043 2.51e-15 ***
clarityVS1    5629.81     459.72  12.246  < 2e-16 ***
clarityVS2    5158.56     452.76  11.394  < 2e-16 ***
clarityVVS1   5713.48     477.52  11.965  < 2e-16 ***
clarityVVS2   5639.76     464.90  12.131  < 2e-16 ***
depth          -13.05      31.91  -0.409 0.682618
table          -51.57      25.54  -2.019 0.043761 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1350 on 985 degrees of freedom
Multiple R-squared:  0.8905,    Adjusted R-squared:  0.8889
F-statistic: 572.1 on 14 and 985 DF,  p-value: < 2.2e-16
```

Question 5: Comments on Part 1

1. Price shows a very strong positive correlation with carat. This confirms that the weight of the diamond is a good predictor of diamond price.
2. Both price and carat are highly right-skewed. This suggests that log-transformations will likely improve model assumptions.
3. Most diamonds in our sample have Ideal or Premium cuts, G or E colors, and SI1 or VS2 clarity grade which shows that our sample contains mostly mid to high quality diamonds.
4. ANOVA results confirmed that cut and clarity significantly affect price.

PART 2: Simple Linear Regression

Question 1: Start with one predictor and one response

```r
slr_model <- lm(price ~ carat, data = diamonds_sample)
```

Question 2: Run the model and examine summary statistics

```r
summary(slr_model)
```

```
Call:
lm(formula = price ~ carat, data = diamonds_sample)

Residuals:
     Min       1Q   Median       3Q      Max 
-10255.1   -855.9     -7.8    532.9  10929.3 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -2249.5      102.5  -21.95   <2e-16 ***
carat         7789.2      110.0   70.83   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1650 on 998 degrees of freedom
Multiple R-squared:  0.8341,	Adjusted R-squared:  0.8339 
F-statistic:  5017 on 1 and 998 DF,  p-value: < 2.2e-16
```

```r
# Confidence intervals for coefficients
confint(slr_model)
```

```
                2.5 %    97.5 %
(Intercept) -2450.582 -2048.362
carat        7573.400  8005.003
```

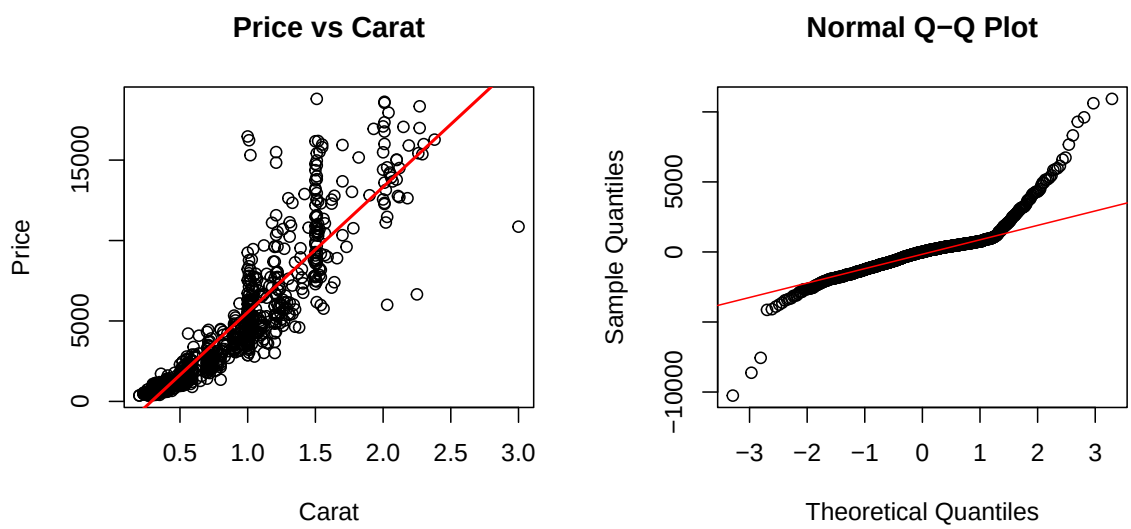
```r
# Create a new data point for prediction
new_data <- data.frame(carat = 0.5)

# Confidence interval for mean response
predict(slr_model, newdata = new_data, interval = "confidence")
```

```
       fit      lwr      upr
1 1645.129 1523.736 1766.521
```

```r
# Prediction interval for individual response
predict(slr_model, newdata = new_data, interval = "prediction")
```

```
       fit       lwr      upr
1 1645.129 -1595.575 4885.832
```

```r
# Scatter plot with regression line
par(mfrow = c(1, 2))
plot(diamonds_sample$carat, diamonds_sample$price,
     xlab = "Carat", ylab = "Price",
     main = "Price vs Carat")
abline(slr_model, col = "red", lwd = 2)

# QQ plot for residuals
qqnorm(slr_model$residuals)
qqline(slr_model$residuals, col = "red")
```

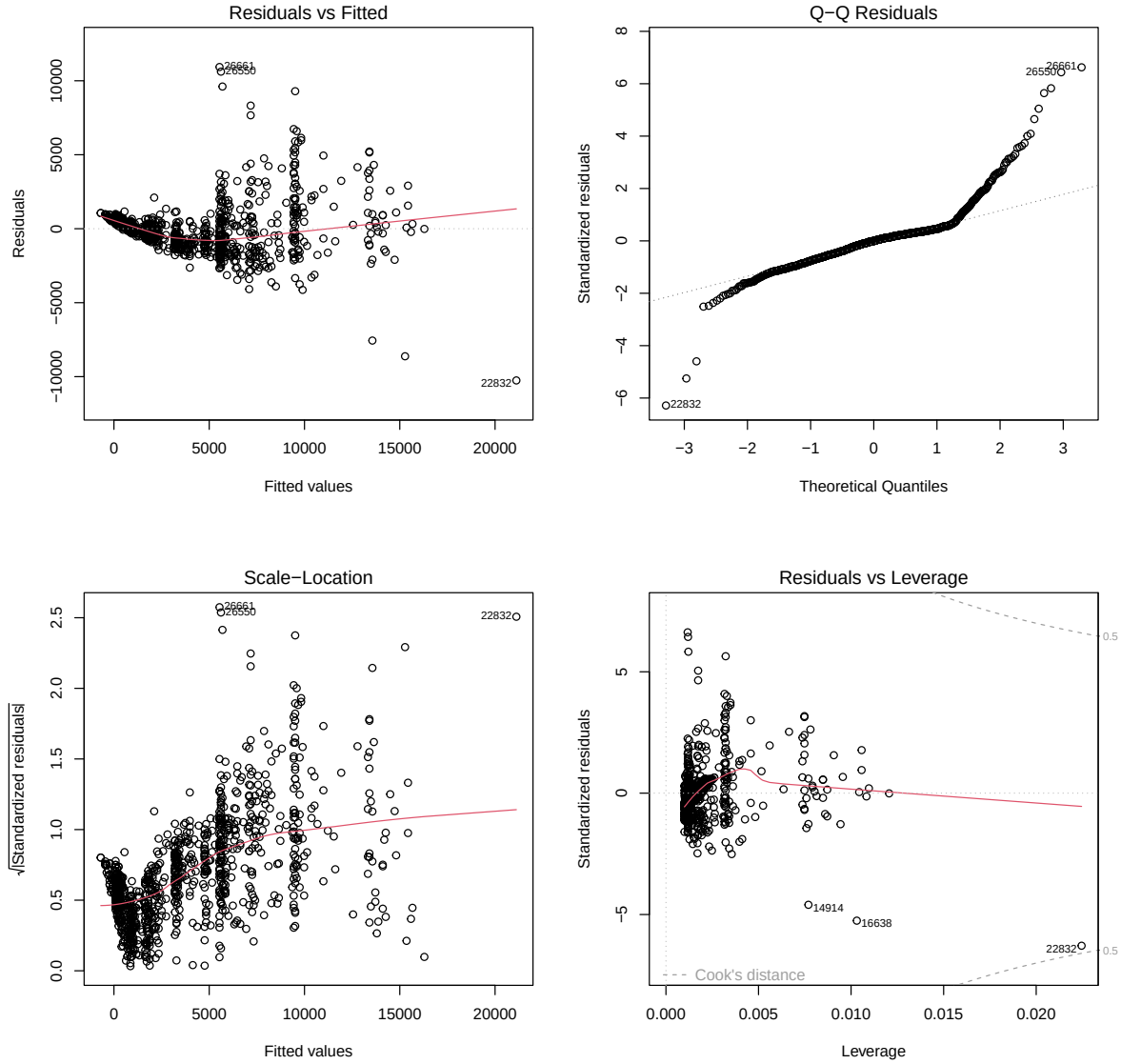


- **Carat Coefficient Estimate:** The slope (β_1) of **7789.2** indicates that a one 'carat' increase is associated with a predicted increase in 'price' of $\$\mathbf{7,789.2}$.
- **Hypothesis Test:** The overall F-statistic is highly significant ($\mathbf{P < 2e\text{-}16}$), meaning the model is statistically valid and the 'carat' coefficient is significantly non-zero.
- **Multiple** R^2: The value of **0.8341** shows that **83.41**% of the total variability in 'price' is explained by the 'carat' predictor.

- **Adjusted R^2**: The value of **0.8339** is negligibly lower than Multiple R^2, indicating that the model's fit is good even after accounting for the use of one predictor.
- **Residual Standard Error (RSE):** The RSE of $\$\mathbf{1,650}$ is large, meaning that the average prediction error is high.
- **Confidence Interval (CI):** This interval provides the likely range for the **true mean price** of all diamonds sharing a specific 'carat' weight.
- **Prediction Interval (PI):** This interval provides the wider, likely range for the 'price' of a **single, new diamond** of a specific 'carat' weight.
- **Plot & Assumptions:** The scatterplot and residual diagnostics visually confirm a **curved, fanned-out relationship**, violating the core assumptions of linearity and homoscedasticity.

Question 3: Test assumptions and apply transformations

```r
# Testing Assumptions
par(mfrow = c(2, 2))
plot(slr_model)
```




```
# Applying Log transformation to both response and predictor
diamonds_sample$log_price <- log(diamonds_sample$price)
diamonds_sample$log_carat <- log(diamonds_sample$carat)
```

Assumption Assessment and Transformation Decision: The diagnostics plots from the simple model show clear **non-linearity** (a curve in the Residuals vs Fitted plot) and severe **heteroscedasticity** (a funnel shape). The Normal Q-Q plot also indicates highly non-normal residuals due to the right skew of 'price'.

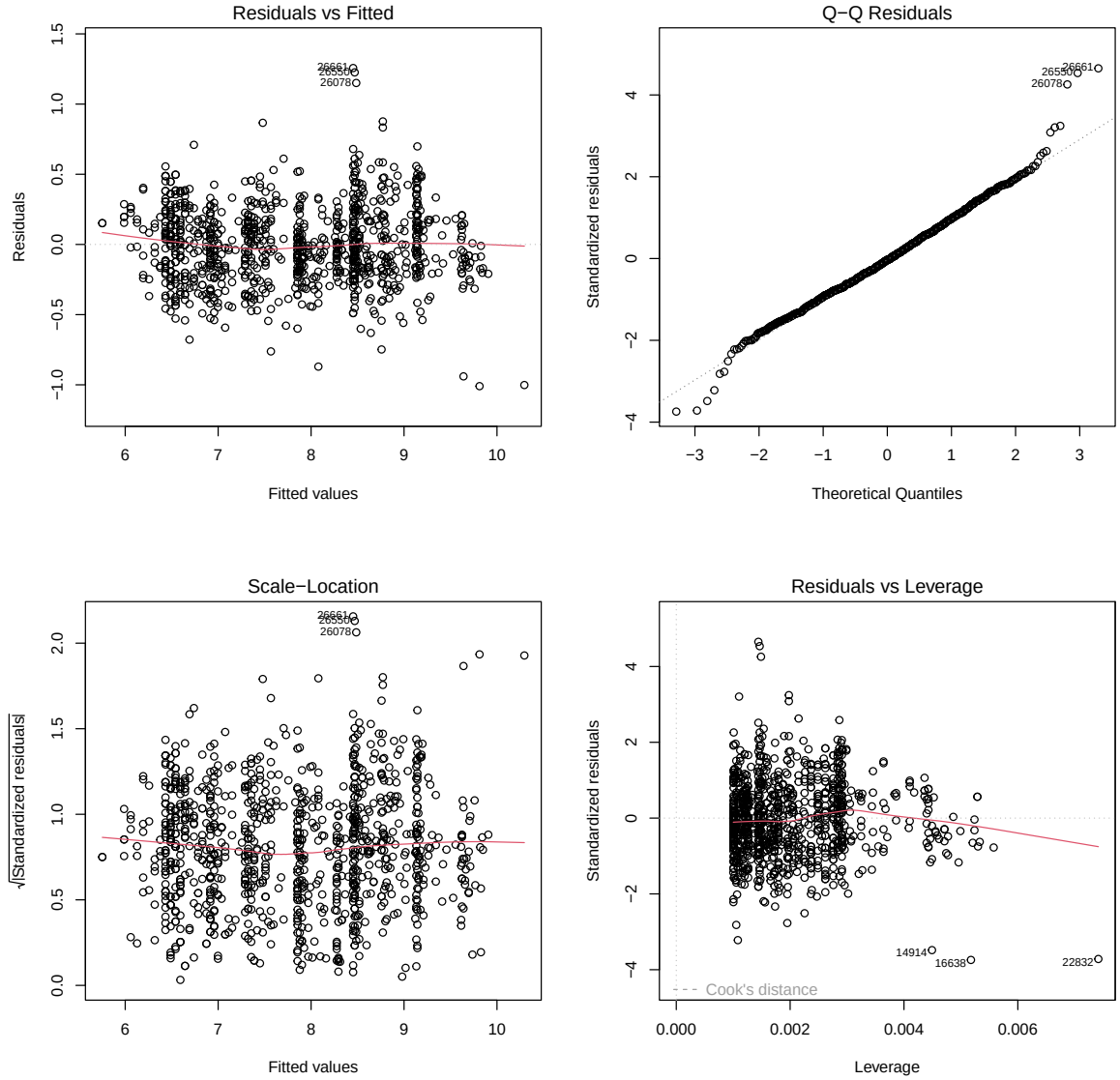
To correct these violations and validate the model's assumptions, we apply a **log-log transformation** to both the response variable (price) and the predictor (carat).

Question 4: Summary of Transformed Variables and Changes

The model is refitted as $\log(\text{price}) \sim \log(\text{carat})$.

```
# New model with transformed response
slr_transformed <- lm(log_price ~ log_carat, data = diamonds_sample)
```

```
# Diagnostic plots for transformed model
par(mfrow = c(2, 2))
plot(slr_transformed)
```




```r
summary(slr_transformed)
```

```
Call:
lm(formula = log_price ~ log_carat, data = diamonds_sample)

Residuals:
     Min       1Q   Median       3Q      Max
```

```
-1.00947 -0.18659 -0.00921  0.17056  1.25679

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.45244    0.01027   823.0   <2e-16 ***
log_carat    1.67690    0.01454   115.3   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2704 on 998 degrees of freedom
Multiple R-squared:  0.9302,    Adjusted R-squared:  0.9302
F-statistic: 1.331e+04 on 1 and 998 DF,  p-value: < 2.2e-16
```

Changes Noted: The diagnostic plots for the transformed model show vast improvement: the **Residuals vs. Fitted** plot now shows a random scatter, confirming **linearity and homoscedasticity** are largely restored. The **Normal Q-Q plot** shows points falling much closer to the theoretical line, indicating normality of residuals is improved. The Adjusted R^2 is higher, and the Residual Standard Error is reported on the log scale, reflecting the significant reduction in scatter and error compared to the raw price scale.

Question 5: Add other variables and assess improvement

The best model includes all of the variables. Each of the variables that was added to the model increased the Adjusted R^2.

Question 6: Comments on Part 2

1. The log-log transformation was necessary to satisfy the regression assumptions as the original model showed violations of linearity and homosecdasticity.
2. Adding all of the variables in the dataset improved the Adjusted R^2 from 0.93 to 0.98.
3. Carat is still the most important predictor in the dataset and explains about 93% of price variation of its own when log transformed.

PART 3: Multiple Linear Regression

Question 1: Best model selection

```r
# Based on Part 2, our best model includes all variables with log(price)
summary(best_model)
```

```
Call:
lm(formula = log_price ~ log_carat + depth + table + cut + color +
    clarity + x + y + z, data = diamonds_sample)

Residuals:
     Min       1Q   Median       3Q      Max
-0.41991 -0.08965  0.00106  0.08433  0.52588

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.1939093  0.4391142  16.383  < 2e-16 ***
log_carat     1.6373959  0.0729726  22.438  < 2e-16 ***
depth        -0.0006526  0.0035031  -0.186 0.852253
table        -0.0033757  0.0024661  -1.369 0.171354
cutGood       0.0885617  0.0280910   3.153 0.001667 **
cutIdeal      0.1488318  0.0276717   5.378 9.40e-08 ***
cutPremium    0.1363247  0.0269664   5.055 5.13e-07 ***
cutVery Good  0.1017617  0.0273493   3.721 0.000210 ***
colorE       -0.0506658  0.0148795  -3.405 0.000688 ***
colorF       -0.1023908  0.0150620  -6.798 1.84e-11 ***
colorG       -0.1617233  0.0143144 -11.298  < 2e-16 ***
colorH       -0.2756313  0.0157739 -17.474  < 2e-16 ***
colorI       -0.4093315  0.0172088 -23.786  < 2e-16 ***
colorJ       -0.5348651  0.0203686 -26.259  < 2e-16 ***
clarityIF     1.2151340  0.0485232  25.042  < 2e-16 ***
claritySI1    0.6470645  0.0433512  14.926  < 2e-16 ***
claritySI2    0.4915751  0.0435140  11.297  < 2e-16 ***
clarityVS1    0.8747369  0.0443729  19.713  < 2e-16 ***
clarityVS2    0.8066861  0.0436226  18.492  < 2e-16 ***
clarityVVS1   1.0652439  0.0458239  23.246  < 2e-16 ***
clarityVVS2   0.9671601  0.0447724  21.602  < 2e-16 ***
x             0.0779276  0.0967074   0.806 0.420550
y             0.0292426  0.0953298   0.307 0.759097
z             0.0423896  0.0252469   1.679 0.093471 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.129 on 976 degrees of freedom
Multiple R-squared:  0.9845,    Adjusted R-squared:  0.9841
F-statistic:  2692 on 23 and 976 DF,  p-value: < 2.2e-16
```

Observations from summary: - Adjusted R-squared: 0.9841 (explains 98.41% of variance) -

Predictors depth, table, x, y, and z are not significant - Model is highly statistically significant

```r
# Backward elimination using AIC
backward_aic <- step(best_model, direction = "backward", trace = FALSE)
summary(backward_aic)
```

```
Call:
lm(formula = log_price ~ log_carat + cut + color + clarity +
    x + z, data = diamonds_sample)

Residuals:
     Min       1Q   Median       3Q      Max
-0.42146 -0.09105  0.00159  0.08191  0.52360

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.94734    0.23777  29.219  < 2e-16 ***
log_carat     1.63506    0.06893  23.722  < 2e-16 ***
cutGood       0.09029    0.02789   3.238 0.001246 **
cutIdeal      0.16080    0.02609   6.163 1.04e-09 ***
cutPremium    0.13895    0.02668   5.208 2.33e-07 ***
cutVery Good  0.10885    0.02636   4.129 3.96e-05 ***
colorE       -0.05160    0.01484  -3.477 0.000529 ***
colorF       -0.10313    0.01499  -6.878 1.08e-11 ***
colorG       -0.16196    0.01429 -11.331  < 2e-16 ***
colorH       -0.27517    0.01570 -17.532  < 2e-16 ***
colorI       -0.41028    0.01718 -23.886  < 2e-16 ***
colorJ       -0.53449    0.02021 -26.442  < 2e-16 ***
clarityIF     1.21805    0.04826  25.242  < 2e-16 ***
claritySI1    0.64872    0.04315  15.035  < 2e-16 ***
claritySI2    0.49297    0.04339  11.362  < 2e-16 ***
clarityVS1    0.87686    0.04415  19.861  < 2e-16 ***
clarityVS2    0.80823    0.04347  18.593  < 2e-16 ***
clarityVVS1   1.06731    0.04564  23.384  < 2e-16 ***
clarityVVS2   0.96884    0.04453  21.757  < 2e-16 ***
x             0.10485    0.03474   3.018 0.002610 **
z             0.04700    0.02365   1.987 0.047163 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1289 on 979 degrees of freedom
```

```
Multiple R-squared:  0.9844,    Adjusted R-squared:  0.9841
F-statistic:  3098 on 20 and 979 DF,  p-value: < 2.2e-16
```

```r
# Backward elimination using BIC criterion
n <- nrow(diamonds_sample)
backward_bic <- step(best_model, direction = "backward", k = log(n), trace = FALSE)
summary(backward_bic)
```

```
Call:
lm(formula = log_price ~ log_carat + cut + color + clarity +
    x, data = diamonds_sample)

Residuals:
     Min       1Q   Median       3Q      Max
-0.42134 -0.09100  0.00110  0.08353  0.51991

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.09086    0.22688  31.254  < 2e-16 ***
log_carat     1.67507    0.06602  25.372  < 2e-16 ***
cutGood       0.08821    0.02791   3.160 0.001623 **
cutIdeal      0.15776    0.02608   6.048 2.08e-09 ***
cutPremium    0.13444    0.02662   5.050 5.28e-07 ***
cutVery Good  0.10580    0.02636   4.014 6.43e-05 ***
colorE       -0.05129    0.01486  -3.451 0.000582 ***
colorF       -0.10314    0.01502  -6.868 1.15e-11 ***
colorG       -0.16161    0.01431 -11.291  < 2e-16 ***
colorH       -0.27574    0.01572 -17.545  < 2e-16 ***
colorI       -0.40958    0.01720 -23.814  < 2e-16 ***
colorJ       -0.53251    0.02022 -26.337  < 2e-16 ***
clarityIF     1.21665    0.04832  25.177  < 2e-16 ***
claritySI1    0.64698    0.04320  14.975  < 2e-16 ***
claritySI2    0.49003    0.04343  11.284  < 2e-16 ***
clarityVS1    0.87514    0.04421  19.796  < 2e-16 ***
clarityVS2    0.80657    0.04353  18.530  < 2e-16 ***
clarityVVS1   1.06607    0.04571  23.324  < 2e-16 ***
clarityVVS2   0.96688    0.04459  21.685  < 2e-16 ***
x             0.11243    0.03458   3.251 0.001188 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1291 on 980 degrees of freedom
Multiple R-squared:  0.9844,    Adjusted R-squared:  0.9841
F-statistic:  3251 on 19 and 980 DF,  p-value: < 2.2e-16
```

```r
# Stepwise regression using AIC on null model
null_model <- lm(log_price ~ 1, data = diamonds_sample)
stepwise_aic <- step(null_model,
                     scope = list(lower = null_model, upper = best_model),
                     direction = "both", trace = FALSE)
summary(stepwise_aic)
```

```
Call:
lm(formula = log_price ~ log_carat + clarity + color + cut +
    x + z, data = diamonds_sample)

Residuals:
     Min       1Q   Median       3Q      Max
-0.42146 -0.09105  0.00159  0.08191  0.52360

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.94734    0.23777  29.219  < 2e-16 ***
log_carat     1.63506    0.06893  23.722  < 2e-16 ***
clarityIF     1.21805    0.04826  25.242  < 2e-16 ***
claritySI1    0.64872    0.04315  15.035  < 2e-16 ***
claritySI2    0.49297    0.04339  11.362  < 2e-16 ***
clarityVS1    0.87686    0.04415  19.861  < 2e-16 ***
clarityVS2    0.80823    0.04347  18.593  < 2e-16 ***
clarityVVS1   1.06731    0.04564  23.384  < 2e-16 ***
clarityVVS2   0.96884    0.04453  21.757  < 2e-16 ***
colorE       -0.05160    0.01484  -3.477 0.000529 ***
colorF       -0.10313    0.01499  -6.878 1.08e-11 ***
colorG       -0.16196    0.01429 -11.331  < 2e-16 ***
colorH       -0.27517    0.01570 -17.532  < 2e-16 ***
colorI       -0.41028    0.01718 -23.886  < 2e-16 ***
colorJ       -0.53449    0.02021 -26.442  < 2e-16 ***
cutGood       0.09029    0.02789   3.238 0.001246 **
cutIdeal      0.16080    0.02609   6.163 1.04e-09 ***
cutPremium    0.13895    0.02668   5.208 2.33e-07 ***
cutVery Good  0.10885    0.02636   4.129 3.96e-05 ***
x             0.10485    0.03474   3.018 0.002610 **
```

```
z             0.04700    0.02365   1.987 0.047163 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1289 on 979 degrees of freedom
Multiple R-squared:  0.9844,    Adjusted R-squared:  0.9841
F-statistic:  3098 on 20 and 979 DF,  p-value: < 2.2e-16
```

Observations: - Full model Adjusted R^2: 0.9841, explaining 98.41% of variance in log(price) - All selection methods do not retain depth, table, and y - AIC and Stepwise AIC retain both x and z while BIC only retains x. The models explain approximately 98% of variance in log(price).

Question 2: Detect multicollinearity using VIF

```r
# Calculate VIF for the best model
vif_values <- vif(best_model)
print(vif_values)
```

```
                GVIF Df GVIF^(1/(2*Df))
log_carat 110.768682  1       10.524670
depth       1.629992  1        1.276712
table       1.713255  1        1.308914
cut         2.596736  4        1.126687
color       1.301737  6        1.022218
clarity     1.504625  7        1.029612
x         710.338816  1       26.652182
y         683.182432  1       26.137759
z          18.876703  1        4.344733
```

```r
# Identify bad VIF values (> 10)
high_vif <- vif_values[vif_values > 10]
print("Variables with VIF > 10:")
```

```
[1] "Variables with VIF > 10:"
```

```r
print(high_vif)
```

```
[1] 110.76868 710.33882 683.18243  18.87670  10.52467  26.65218  26.13776
```

Multicollinearity Assessment: - x, y, z, and carat have extremely high VIF values, which signals their multicollinearity.

```r
# Reduced Model without depth, table, x, y, and z.
reduced_model <- lm(log_price ~ log_carat + cut + color + clarity,
                    data = diamonds_sample)
summary(reduced_model)
```

```
Call:
lm(formula = log_price ~ log_carat + cut + color + clarity, data = diamonds_sample)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.43461 -0.09163  0.00093  0.08040  0.51454 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   7.81352    0.04570 170.977  < 2e-16 ***
log_carat     1.88816    0.00797 236.911  < 2e-16 ***
cutGood       0.10197    0.02772   3.678 0.000248 ***
cutIdeal      0.17670    0.02555   6.917 8.34e-12 ***
cutPremium    0.15768    0.02577   6.119 1.36e-09 ***
cutVery Good  0.12234    0.02599   4.707 2.87e-06 ***
colorE       -0.05229    0.01493  -3.502 0.000483 ***
colorF       -0.10608    0.01506  -7.043 3.55e-12 ***
colorG       -0.16194    0.01438 -11.259  < 2e-16 ***
colorH       -0.27166    0.01574 -17.257  < 2e-16 ***
colorI       -0.40803    0.01728 -23.619  < 2e-16 ***
colorJ       -0.52996    0.02030 -26.103  < 2e-16 ***
clarityIF     1.20448    0.04841  24.880  < 2e-16 ***
claritySI1    0.63135    0.04314  14.634  < 2e-16 ***
claritySI2    0.47755    0.04347  10.986  < 2e-16 ***
clarityVS1    0.86228    0.04424  19.489  < 2e-16 ***
clarityVS2    0.79413    0.04357  18.227  < 2e-16 ***
clarityVVS1   1.05493    0.04580  23.033  < 2e-16 ***
clarityVVS2   0.95335    0.04461  21.371  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1297 on 981 degrees of freedom
Multiple R-squared:  0.9842,	Adjusted R-squared:  0.9839 
```

```
F-statistic:  3398 on 18 and 981 DF,  p-value: < 2.2e-16
```

```r
# VIF for Reduced Model
vif_reduced <- vif(reduced_model)
print("VIF values for reduced model:")
```

```
[1] "VIF values for reduced model:"
```

```r
print(vif_reduced)
```

```
              GVIF Df GVIF^(1/(2*Df))
log_carat 1.305911  1        1.142765
cut       1.183373  4        1.021269
color     1.225603  6        1.017097
clarity   1.432785  7        1.026020
```

All VIF values in the reduced model are below 5, indicating no multicollinearity.

Question 3: Confidence and Prediction Intervals

```r
# New data point for prediction
new_observation <- data.frame(
  log_carat = log(0.7),
  cut = "Ideal",
  color = "G",
  clarity = "VS2"
)

# Confidence interval for mean response (log scale)
ci_mean_log <- predict(reduced_model, newdata = new_observation,
                       interval = "confidence", level = 0.95)
print("95% CI for mean log(price):")
```

```
[1] "95% CI for mean log(price):"
```

```r
print(ci_mean_log)
```

```
       fit      lwr      upr
1 7.948949 7.923554 7.974343
```

```r
# Transform back to original price scale
ci_mean_price <- exp(ci_mean_log)
print("95% CI for mean price ($):")
```

```
[1] "95% CI for mean price ($):"
```

```r
print(ci_mean_price)
```

```
       fit     lwr      upr
1 2832.596 2761.57 2905.449
```

```r
# Prediction interval for individual response (log scale)
pi_individual_log <- predict(reduced_model, newdata = new_observation,
                             interval = "prediction", level = 0.95)
print("95% PI for individual log(price):")
```

```
[1] "95% PI for individual log(price):"
```

```r
print(pi_individual_log)
```

```
       fit      lwr      upr
1 7.948949 7.693102 8.204796
```

```r
# Transform back to original price scale
pi_individual_price <- exp(pi_individual_log)
print("95% PI for individual price ($):")
```

```
[1] "95% PI for individual price ($):"
```

```r
print(pi_individual_price)
```

```
       fit      lwr      upr
1 2832.596 2193.167 3658.454
```

Explanation: - The confidence interval provides the range for the mean price of all diamonds with these characteristics - The prediction interval provides the range for a single diamond's price with these characteristics - The prediction interval is wider than the confidence interval, as it accounts for individual variation

Question 4: Summary Report

This regression analysis successfully developed a predictive model for diamond prices using physical characteristics and quality measures from a sample of 1,000 diamonds.

Key Findings:

1. **Model Performance**:
   - Full model with depth, table, x, y, and z: Adjusted $R^2 = 0.9841$
   - Reduced model without depth, table, x, y, and z: Adjusted $R^2 = 0.9839$
   - Both models are a good fit
   - The reduced model is preferred due to resolved multicollinearity and Stepwise AIC selection.
2. **Significant Predictors (in order of importance)**:
   - Carat (log-transformed): Strongest predictor with a 1% increase in carat giving about a 1.7% increase in price.
   - Clarity: Higher clarity grades significantly increase price.
   - Color: Better color grades means higher prices.
   - Cut: Ideal cuts are valued higher than fair cuts.
3. **Multicollinearity**:
   - The original model had severe multicollinearity with severe VIF scores greater than 10 for x, y, z and log_carat.
   - This is because x, y, and z are direct measures of size and are highly correlated with carat.
   - We decided to remove x, y, and z to retain log_carat as the main size measure.
   - This resulted in all VIF values being less than 5.
4. **Model Assumptions:**
   - Log-log transformation was essential for satisfying linear regression assumption requirements.
   - The residual plots and tests confirm assumptions were met.
5. **Practical Interpretation:**
   - For a 0.7 carat, Ideal cut, G color, VS2 clarity diamond:
   - 95% CI for mean price: Provides expected average market value
   - 95% PI for individual price: Accounts for natural variation in individual diamonds